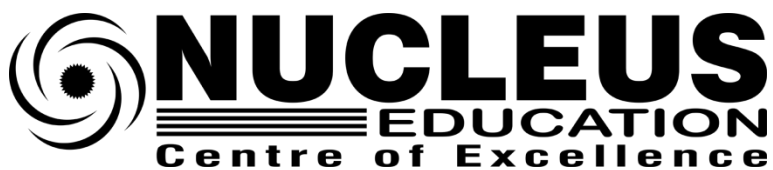


IIT MATHEMATICS

SOLUTIONS OF TRIANGLE



INDEX

S.No.	Contents	Page No.
SOLUTIONS OF TRIANGLE		
1.	Theory	01-16
2.	Exercise # 1 (A) ELEMENTRY	17-19
3.	Exercise # 1 (B)	20-21
4.	Exercise # 2	22
5.	Exercise # 3 (JM)	23
6.	Exercise # 4 (JA)	24-26
7.	Exercise # 5	27-29
8.	Exercise # 6	30
9.	Answer key	31

RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
Gen. Category - 2014

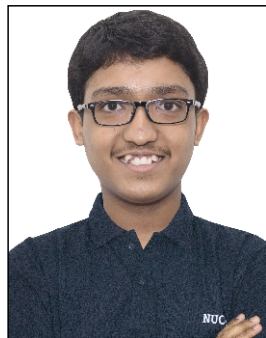


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NISHIT AGARWAL
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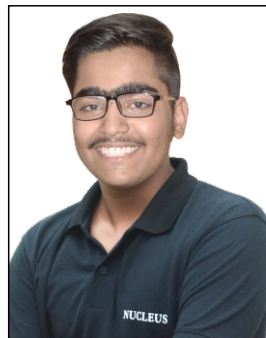


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AMEY GUPTA
Gen. Category - 2014

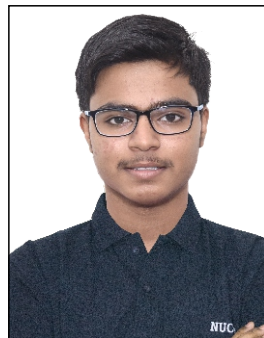
JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



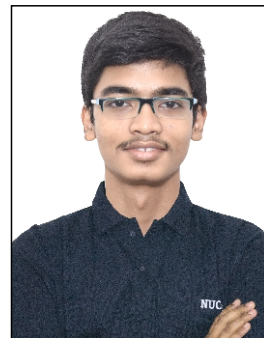
100 Percentile
(Physics)
DAKSH KHANDLWAL
2020



100 Percentile
(Maths & Physics)
VAIBHAV SAHA
2020



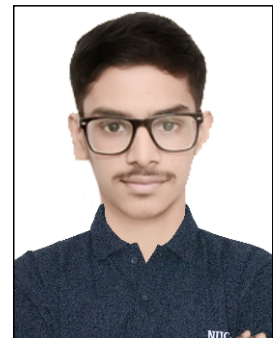
100 Percentile
(Physics)
ANISH MOHAN
2020



100 Percentile
(Maths)
ARCHIT PATNAIK
2020

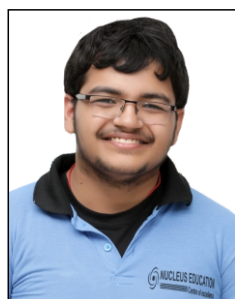


100 Percentile
(Maths)
SWAPNIL YASASVI
2020

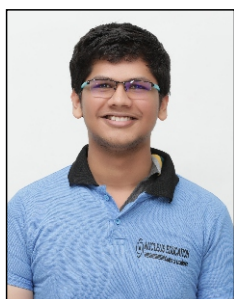


100 Percentile
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PARSHANT ARORA
2020

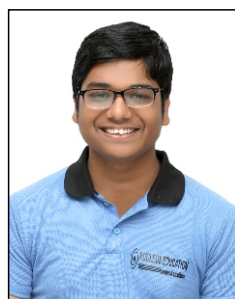
JEE ADVANCED RESULTS OF NUCLEUS EDUCATION



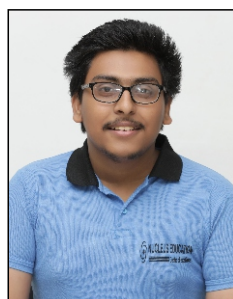
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LAKSHAY SHARMA
Gen. Category - 2017



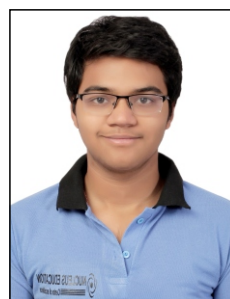
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YATEESH AGRAWAL
Gen. Category - 2017



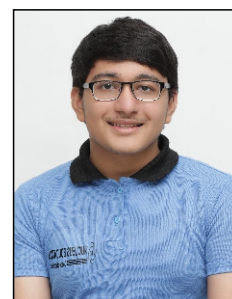
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ABHEY GOYAL
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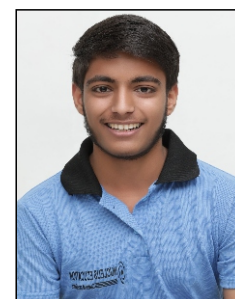
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TUSHAR GAUTAM
Gen. Category - 2017



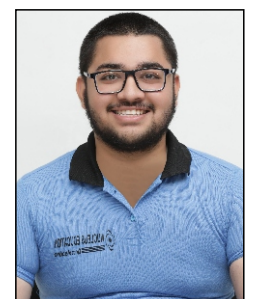
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PIYUSH TIBAREWAL
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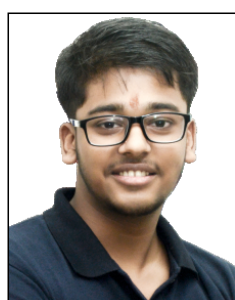
AIR-42
SATVIK
Gen. Category - 2017



AIR-66
MAYANK DUBEY
Gen. Category - 2017



AIR-98
HRITHIK
Gen. Category - 2017



AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



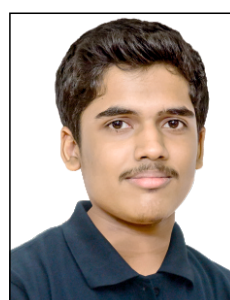
AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
Gen. Category - 2018



AIR-61
SIDDHANT CHOUDHARY
Gen. Category - 2018



AIR-67
ANISWAR S KRISHNAN
Gen. Category - 2018
DLP



AIR-78
AAYUSH KADAM
Gen. Category - 2018



AIR-61
SARTHAK BEHERA
Gen. Category - 2018
*SDCCP



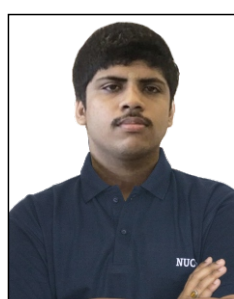
AIR-91
ANDREWS G. VARGHESE
Gen. Category - 2018
*SDCCP



AIR-2
HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



AIR-19
VIBHAV AGGARWAL
Gen. Category - 2019



AIR-33
S. PRAJEETH
Gen. Category - 2019
*SDCCP



AIR-48
SOHAM MISTRI
Gen. Category - 2019
DLP



AIR-51
SAYANTAN DHAR
Gen. Category - 2019
DLP



AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
DLP



AIR-86
SATVIK JAIN
Gen. Category - 2019

SOLUTIONS OF TRIANGLE

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a $\triangle ABC$, the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE :

In any triangle ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$$

where R is circumradius and Δ is area of triangle.

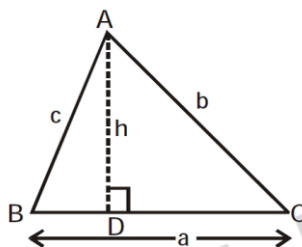


Illustration 1 : Angles of a triangle are in 4 : 1 : 1 ratio. The ratio between its greatest side and perimeter is

- (A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Solution :

Angles are in ratio 4 : 1 : 1.

$\Rightarrow$ angles are $120^\circ, 30^\circ, 30^\circ$.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side.

Now from sine formula $\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$

$$\Rightarrow \frac{a}{\sqrt{3}/2} = \frac{b}{1/2} = \frac{c}{1/2}$$

$$\Rightarrow \frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1} = k \text{ (say)}$$

then $a = \sqrt{3}k$, perimeter = $(2 + \sqrt{3})k$

$$\therefore \text{required ratio} = \frac{\sqrt{3}k}{(2 + \sqrt{3})k} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

Ans. (B)

Illustration 2 : In triangle ABC, if $b = 3$, $c = 4$ and $\angle B = \pi/3$, then number of such triangles is -

- (A) 1 (B) 2 (C) 0 (D) infinite

Solution :

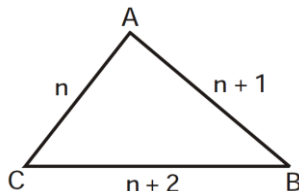
Using sine formulae $\frac{\sin B}{b} = \frac{\sin C}{c}$

$$\Rightarrow \frac{\sin \pi/3}{3} = \frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6} = \frac{\sin C}{4} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1 \text{ which is not possible.}$$

Hence there exist no triangle with given elements.

Ans.(C)

Illustration 3 : The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.



Solution :

Let the sides be $n, n+1, n+2$ cms.

i.e. $AC = n, AB = n+1, BC = n+2$

Smallest angle is B and largest one is A.

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180^\circ$

$$\Rightarrow 3\angle B + \angle C = 180^\circ \Rightarrow \angle C = 180^\circ - 3\angle B$$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin 3B}{n+1}$$

(i) (ii) (iii)

from (i) and (ii);

$$\frac{2\sin B \cos B}{n+2} = \frac{\sin B}{n} \Rightarrow \cos B = \frac{n+2}{2n} \dots\dots\dots(\text{iv})$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3\sin B - 4\sin^3 B}{n+1}$$

$$\Rightarrow \frac{\sin B}{n} = \frac{\sin B(3-4\sin^2 B)}{n+1}$$

$$\Rightarrow \frac{n+1}{n} = 3-4(1-\cos^2 B) \dots\dots\dots(\text{v})$$

from (iv) and (v), we get

$$\frac{n+1}{n} = -1 + 4\left(\frac{n+2}{2n}\right)^2$$

$$\Rightarrow \frac{n+1}{n} + 1 = \left(\frac{n^2 + 4n + 4}{n^2}\right)$$

$$\Rightarrow \frac{2n+1}{n} = \frac{n^2 + 4n + 4}{n^2} \Rightarrow 2n^2 + n = n^2 + 4n + 4$$

$$\Rightarrow n^2 - 3n - 4 = 0 \Rightarrow (n-4)(n+1) = 0$$

$$n = 4 \text{ or } -1$$

where $n \neq -1$

$\therefore n = 4$. Hence the sides are 4, 5, 6

Ans.

Do yourself - 1 :

- (i) If in a $\triangle ABC$, $\angle A = \frac{\pi}{6}$ and $b : c = 2 : \sqrt{3}$, find $\angle B$.
- (ii) Show that, in any $\triangle ABC$: $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) = 0$.
- (iii) If in a $\triangle ABC$, $\frac{\sin A}{\sin C} = \frac{\sin(A - B)}{\sin(B - C)}$, show that a^2, b^2, c^2 are in A.P.

2. COSINE FORMULAE :

$$(a) \quad \cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad (b) \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca} \quad (c) \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{or } a^2 = b^2 + c^2 - 2bc \cos A$$

Illustration 4 : In a triangle ABC, if $B = 30^\circ$ and $c = \sqrt{3} b$, then A can be equal to -
(A) 45° (B) 60° (C) 90° (D) 120°

Solution : We have $\cos B = \frac{c^2 + a^2 - b^2}{2ca} \Rightarrow \frac{\sqrt{3}}{2} = \frac{3b^2 + a^2 - b^2}{2 \times \sqrt{3}b \times a}$
 $\Rightarrow a^2 - 3ab + 2b^2 = 0 \Rightarrow (a - 2b)(a - b) = 0$
 $\Rightarrow$ Either $a = b \Rightarrow A = 30^\circ$
 or $a = 2b$
 $\Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90^\circ$. **Ans. (C)**

Illustration 5 : In a triangle ABC, $(a^2 - b^2 - c^2) \tan A + (a^2 - b^2 + c^2) \tan B$ is equal to -
(A) $(a^2 + b^2 - c^2) \tan C$ (B) $(a^2 + b^2 + c^2) \tan C$
(C) $(b^2 + c^2 - a^2) \tan C$ (D) none of these

Solution : Using cosine law :
 The given expression is equal to $-2bc \cos A \tan A + 2ac \cos B \tan B$
 $= 2abc \left(-\frac{\sin A}{a} + \frac{\sin B}{b} \right) = 0$ **Ans. (D)**

Do yourself - 2 :

- (i) If $a : b : c = 4 : 5 : 6$, then show that $\angle C = 2\angle A$.
- (ii) In any $\triangle ABC$, prove that

$$(a) \quad \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

$$(b) \quad \frac{b^2}{a} \cos A + \frac{c^2}{b} \cos B + \frac{a^2}{c} \cos C = \frac{a^4 + b^4 + c^4}{2abc}$$

3. PROJECTION FORMULAE :

- (a) $b \cos C + c \cos B = a$
 (b) $c \cos A + a \cos C = b$
 (c) $a \cos B + b \cos A = c$

Illustration 6 : In a ΔABC , $c \cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution : Here, $\frac{c}{2}(1 + \cos A) + \frac{a}{2}(1 + \cos C) = \frac{3b}{2}$
 $\Rightarrow a + c + (c \cos A + a \cos C) = 3b$
 $\Rightarrow a + c + b = 3b$ {using projection formula}
 $\Rightarrow a + c = 2b$
 which shows a, b, c are in A.P.

Do yourself - 3 :

(i) In a ΔABC , if $\angle A = \frac{\pi}{4}$, $\angle B = \frac{5\pi}{12}$, show that $a + c\sqrt{2} = 2b$.

(ii) In a ΔABC , prove that :

(a) $b(a \cos C - c \cos A) = a^2 - c^2$

(b) $2\left(b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2}\right) = a + b + c$

4. NAPIER'S ANALOGY (TANGENT RULE) :

(a) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$

(b) $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2}$

(c) $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}$

Illustration 7 : In a ΔABC , the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

Solution : Here, $\tan\left(\frac{A-B}{2}\right) = \frac{1}{3} \tan\left(\frac{A+B}{2}\right)$ (i)

using Napier's analogy, $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right)$ (ii)

from (i) & (ii) ;

$$\frac{1}{3} \tan\left(\frac{A+B}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right) \Rightarrow \frac{1}{3} \cot\left(\frac{C}{2}\right) = \left(\frac{a-b}{a+b}\right) \cot\left(\frac{C}{2}\right)$$

$$\left\{ \text{as } A + B + C = \pi \therefore \tan\left(\frac{A+B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cot \frac{C}{2} \right\}$$

$$\Rightarrow \frac{a-b}{a+b} = \frac{1}{3} \text{ or } 3a - 3b = a + b$$

$$2a = 4b \text{ or } \frac{a}{b} = \frac{2}{1} \Rightarrow \frac{b}{a} = \frac{1}{2}$$

Thus the ratio of the sides opposite to the angles is $b : a = 1 : 2$.

Ans.

Do yourself - 4 :

(i) In any $\triangle ABC$, prove that $\frac{b-c}{b+c} = \frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$

(ii) If $\triangle ABC$ is right angled at C, prove that :

(a) $\tan \frac{A}{2} = \sqrt{\frac{c-b}{c+b}}$

(b) $\sin(A-B) = \frac{a^2-b^2}{a^2+b^2}$

5. HALF ANGLE FORMULAE :

$s = \frac{a+b+c}{2}$ = semi-perimeter of triangle.

(a) (i) $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$ (ii) $\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$ (iii) $\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(b) (i) $\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$ (ii) $\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$ (iii) $\cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

(c) (i) $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ (ii) $\tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$ (iii) $\tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$
 $= \frac{\Delta}{s(s-a)}$ $= \frac{\Delta}{s(s-b)}$ $= \frac{\Delta}{s(s-c)}$

(d) **Area of Triangle**

$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C = \frac{1}{2} ap_1 = \frac{1}{2} bp_2 = \frac{1}{2} cp_3$,

where p_1, p_2, p_3 are altitudes from vertices A, B, C respectively.

Illustration 8 : If in a triangle ABC, CD is the angle bisector of the angle ACB, then CD is equal to-

(A) $\frac{a+b}{2ab} \cos \frac{C}{2}$ (B) $\frac{2ab}{a+b} \sin \frac{C}{2}$ (C) $\frac{2ab}{a+b} \cos \frac{C}{2}$ (D) $\frac{b \sin \angle DAC}{\sin(B+C/2)}$

Solution :

$\Delta CAB = \Delta CAD + \Delta CDB$

$\Rightarrow \frac{1}{2} ab \sin C = \frac{1}{2} b \cdot CD \cdot \sin \left(\frac{C}{2}\right) + \frac{1}{2} a \cdot CD \sin \left(\frac{C}{2}\right)$

$\Rightarrow CD (a+b) \sin \left(\frac{C}{2}\right) = ab \left(2 \sin \left(\frac{C}{2}\right) \cos \left(\frac{C}{2}\right)\right)$

So $CD = \frac{2ab \cos(C/2)}{(a+b)}$

and in $\triangle CAD$, $\frac{CD}{\sin \angle DAC} = \frac{b}{\sin \angle CDA}$ (by sine rule)

$\Rightarrow CD = \frac{b \sin \angle DAC}{\sin(B+C/2)}$

Ans.(C,D)

Illustration 9 : If Δ is the area and $2s$ the sum of the sides of a triangle, then show $\Delta \leq \frac{s^2}{3\sqrt{3}}$.

Solution : We have, $2s = a + b + c$, $\Delta^2 = s(s-a)(s-b)(s-c)$

Now, A.M. $\geq$ G.M.

$$\frac{(s-a) + (s-b) + (s-c)}{3} \geq \{(s-a)(s-b)(s-c)\}^{1/3}$$

$$\text{or } \frac{3s - 2s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{\Delta^2}{s} \leq \frac{s^3}{27} \Rightarrow \Delta \leq \frac{s^2}{3\sqrt{3}}$$

Ans.

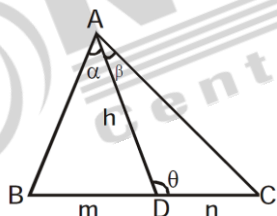
Do yourself - 5 :

(i) Given $a = 6$, $b = 8$, $c = 10$. Find

(a) $\sin A$ (b) $\tan A$ (c) $\sin \frac{A}{2}$ (d) $\cos \frac{A}{2}$ (e) $\tan \frac{A}{2}$ (f) Δ

(ii) Prove that in any ΔABC , $(abc) \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = \Delta^2$

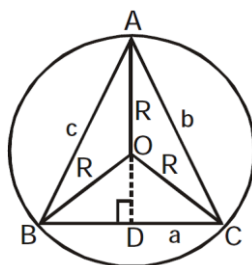
6. **m-n THEOREM :**



$$(m+n) \cot \theta = m \cot \alpha - n \cot \beta$$

$$(m+n) \cot \theta = n \cot B - m \cot C.$$

7. **RADIUS OF THE CIRCUMCIRCLE 'R' :**

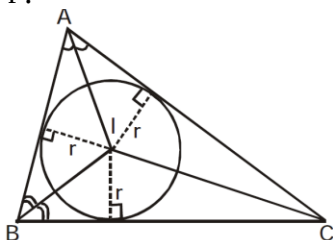


Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

8. RADIUS OF THE INCIRCLE 'r' :

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.



$$r = \frac{\Delta}{s} = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$

$$= a \frac{\sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = b \frac{\sin \frac{A}{2} \sin \frac{C}{2}}{\cos \frac{B}{2}} = c \frac{\sin \frac{B}{2} \sin \frac{A}{2}}{\cos \frac{C}{2}}$$

Illustration 10 : In a triangle ABC, if $a : b : c = 4 : 5 : 6$, then ratio between its circumradius and inradius is-

- (A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Solution :

$$\frac{R}{r} = \frac{abc}{4\Delta} \cdot \frac{\Delta}{s} = \frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)} \dots (i)$$

$$\because a : b : c = 4 : 5 : 6 \Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k(\text{say}) \Rightarrow a = 4k, b = 5k, c = 6k$$

$$\therefore s = \frac{a+b+c}{2} = \frac{15k}{2}, s-a = \frac{7k}{2}, s-b = \frac{5k}{2}, s-c = \frac{3k}{2}$$

$$\text{using (i) in these values } \frac{R}{r} = \frac{(4k)(5k)(6k)}{4\left(\frac{7k}{2}\right)\left(\frac{5k}{2}\right)\left(\frac{3k}{2}\right)} = \frac{16}{7}$$

Ans. (A)

Illustration 11 : If A, B, C are the angles of a triangle, prove that : $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.

Solution :

$$\begin{aligned} \cos A + \cos B + \cos C &= 2 \cos \left(\frac{A+B}{2} \right) \cdot \cos \left(\frac{A-B}{2} \right) + \cos C \\ &= 2 \sin \frac{C}{2} \cdot \cos \left(\frac{A-B}{2} \right) + 1 - 2 \sin^2 \frac{C}{2} = 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \sin \left(\frac{C}{2} \right) \right] \\ &= 1 + 2 \sin \frac{C}{2} \left[\cos \left(\frac{A-B}{2} \right) - \cos \left(\frac{A+B}{2} \right) \right] \left\{ \because \frac{C}{2} = 90^\circ - \left(\frac{A+B}{2} \right) \right\} \\ &= 1 + 2 \sin \frac{C}{2} \cdot 2 \sin \frac{A}{2} \cdot \sin \frac{B}{2} = 1 + 4 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} \\ &= 1 + \frac{r}{R} \quad \{ \text{as, } r = 4R \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} \} \\ &\Rightarrow \cos A + \cos B + \cos C = 1 + \frac{r}{R}. \end{aligned}$$

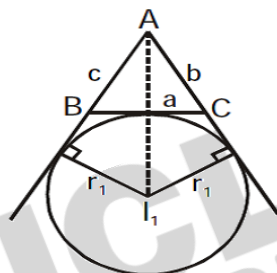
Hence proved.

Do yourself - 6 :

- (i) If in ΔABC , $a = 3$, $b = 4$ and $c = 5$, find
(a) Δ (b) R (c) r
- (ii) In a ΔABC , show that :
(a) $\frac{a^2 - b^2}{c} = 2R \sin(A-B)$ (b) $r \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{\Delta}{4R}$ (c) $a + b + c = \frac{abc}{2Rr}$
- (iii) Let Δ & Δ' denote the areas of a triangle and that of its incircle. Prove that
 $\Delta : \Delta' = \left(\cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2} \right) : \pi$

9. RADII OF THE EX-CIRCLES :

Point of intersection of two external angles and one internal angle bisectors is excentre and perpendicular distance of excentre from any side is called exradius. If r_1 is the radius of escribed circle opposite to $\angle A$ of ΔABC and so on, then –



$$\begin{aligned} \text{(a)} \quad r_1 &= \frac{\Delta}{s-a} = s \tan \frac{A}{2} = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}} \\ \text{(b)} \quad r_2 &= \frac{\Delta}{s-b} = s \tan \frac{B}{2} = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} = \frac{b \cos \frac{A}{2} \cos \frac{C}{2}}{\cos \frac{B}{2}} \\ \text{(c)} \quad r_3 &= \frac{\Delta}{s-c} = s \tan \frac{C}{2} = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} = \frac{c \cos \frac{A}{2} \cos \frac{B}{2}}{\cos \frac{C}{2}} \end{aligned}$$

I_1, I_2 and I_3 are taken as ex-centre opposite to vertex A, B, C respectively.

Illustration 12 : Value of the expression $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3}$ is equal to -

- (A) 1 (B) 2 (C) 3 (D) 0

Solution : $\frac{(b-c)}{r_1} + \frac{(c-a)}{r_2} + \frac{(a-b)}{r_3}$

$$\Rightarrow (b-c) \left(\frac{s-a}{\Delta} \right) + (c-a) \left(\frac{s-b}{\Delta} \right) + (a-b) \left(\frac{s-c}{\Delta} \right)$$

$$\Rightarrow \frac{(s-a)(b-c) + (s-b)(c-a) + (s-c)(a-b)}{\Delta}$$

$$= \frac{s(b-c+c-a+a-b) - [ab-ac+bc-ba+ac-bc]}{\Delta} = \frac{0}{\Delta} = 0$$

$$\text{Thus, } \frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$$

Ans. (D)

Illustration 13 : If $r_1 = r_2 + r_3 + r$, prove that the triangle is right angled.

Solution :

We have, $r_1 - r = r_2 + r_3$

$$\Rightarrow \frac{\Delta}{s-a} - \frac{\Delta}{s} = \frac{\Delta}{s-b} + \frac{\Delta}{s-c} \Rightarrow \frac{s-s+a}{s(s-a)} = \frac{s-c+s-b}{(s-b)(s-c)}$$

$$\Rightarrow \frac{a}{s(s-a)} = \frac{2s-(b+c)}{(s-b)(s-c)} \quad \{as, 2s = a+b+c\}$$

$$\Rightarrow \frac{a}{s(s-a)} = \frac{a}{(s-b)(s-c)} \Rightarrow s^2 - (b+c)s + bc = s^2 - as$$

$$\Rightarrow s(-a+b+c) = bc \Rightarrow \frac{(b+c-a)(a+b+c)}{2} = bc$$

$$\Rightarrow (b+c)^2 - (a)^2 = 2bc \Rightarrow b^2 + c^2 + 2bc - a^2 = 2bc$$

$$\Rightarrow b^2 + c^2 = a^2$$

$$\therefore \angle A = 90^\circ.$$

Ans.

Do yourself - 7 :

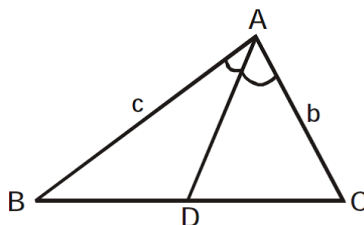
(i) In an equilateral $\triangle ABC$, $R = 2$, find

(a) r (b) r_1 (c) a

(ii) In a $\triangle ABC$, show that

$$(a) r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2 \quad (b) \frac{1}{4} r^2 s^2 \left(\frac{1}{r} - \frac{1}{r_1} \right) \left(\frac{1}{r} - \frac{1}{r_2} \right) \left(\frac{1}{r} - \frac{1}{r_3} \right) = R \quad (c) \sqrt{r r_1 r_2 r_3} = \Delta$$

10. Angle Bisectors & medians :



An angle bisector divides the base in the ratio of corresponding sides.

$$\frac{BD}{CD} = \frac{c}{b} \Rightarrow BD = \frac{ac}{b+c} \text{ and } CD = \frac{ab}{b+c}$$

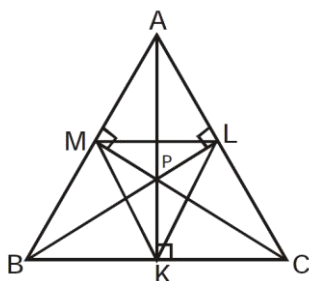
If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} \text{ and } \beta_a = \frac{2bcc \cos \frac{A}{2}}{b+c}$$

$$\text{Note that } m_a^2 + m_b^2 + m_c^2 = \frac{3}{4} (a^2 + b^2 + c^2)$$

11. ORTHOCENTRE :

- (a) Point of intersection of altitudes is orthocentre & the triangle KLM which is formed by joining the feet of the altitudes is called the pedal triangle.



- (b) The distances of the orthocentre from the angular points of the ΔABC are $2R \cos A$, $2R \cos B$, & $2R \cos C$.
(c) The distance of P from sides are $2R \cos B \cos C$, $2R \cos C \cos A$ and $2R \cos A \cos B$.

Do yourself - 8 :

- (i) If x, y, z are the distance of the vertices of ΔABC respectively from the orthocentre, then prove that $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{abc}{xyz}$
(ii) If p_1, p_2, p_3 are respectively the perpendiculars from the vertices of a triangle to the opposite sides, prove that
(a) $p_1 p_2 p_3 = \frac{a^2 b^2 c^2}{8R^3}$ (b) $\Delta = \sqrt{\frac{1}{2} R p_1 p_2 p_3}$
(iii) In a ΔABC , AD is altitude and H is the orthocentre prove that $AH : DH = (\tan B + \tan C) : \tan A$
(iv) In a ΔABC , the lengths of the bisectors of the angle A, B and C are x, y, z respectively. Show that $\frac{1}{x} \cos \frac{A}{2} + \frac{1}{y} \cos \frac{B}{2} + \frac{1}{z} \cos \frac{C}{2} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

12. THE DISTANCES BETWEEN THE SPECIAL POINTS :

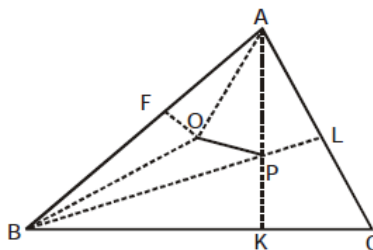
- (a) The distance between circumcentre and orthocentre is $= R \sqrt{1 - 8 \cos A \cos B \cos C}$
(b) The distance between circumcentre and incentre is $= \sqrt{R^2 - 2Rr}$
(c) The distance between incentre and orthocentre is $= \sqrt{2r^2 - 4R^2 \cos A \cos B \cos C}$
(d) The distances between circumcentre & excentres are

$$OI_1 = R \sqrt{1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R^2 + 2Rr_1} \text{ and so on.}$$

Illustration 14 : Prove that the distance between the circumcentre and the orthocentre of a triangle ABC is $R \sqrt{1 - 8 \cos A \cos B \cos C}$.

Solution : Let O and P be the circumcentre and the orthocentre respectively. If OF is the perpendicular to AB, we have $\angle OAF = 90^\circ - \angle AOF = 90^\circ - C$. Also $\angle PAL = 90^\circ - C$. Hence, $\angle OAP = A - \angle OAF - \angle PAL = A - 2(90^\circ - C) = A + 2C - 180^\circ = A + 2C - (A + B + C) = C - B$.
Also $OA = R$ and $PA = 2R \cos A$.

Now in $\triangle AOP$,
 $OP^2 = OA^2 + PA^2 - 2OA \cdot PA \cos OAP$



$$\begin{aligned} &= R^2 + 4R^2 \cos^2 A - 4R^2 \cos A \cos(C - B) \\ &= R^2 + 4R^2 \cos A [\cos A - \cos(C - B)] \\ &= R^2 - 4R^2 \cos A [\cos(B + C) + \cos(C - B)] = R^2 - 8R^2 \cos A \cos B \cos C. \end{aligned}$$

$$\text{Hence } OP = R \sqrt{1 - 8 \cos A \cos B \cos C}.$$

Ans.

13. SOLUTION OF TRIANGLES :

The three sides a, b, c and the three angles A, B, C are called the elements of the triangle ABC . When any three of these six elements (except all the three angles) of a triangle are given, the triangle is known completely; that is the other three elements can be expressed in terms of the given element and can be evaluated. This process is called the solution of triangles.

* If the three sides a, b, c are given, angle A is obtained from $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ or

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}. B \text{ and } C \text{ can be obtained in the similar way.}$$

* If two sides b and c and the included angle A are given, then $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$

gives $\frac{B-C}{2}$. Also $\frac{B+C}{2} = 90^\circ - \frac{A}{2}$, so that B and C can be evaluated. The third side is

$$\text{given by } a = b \frac{\sin A}{\sin B} \text{ or } a^2 = b^2 + c^2 - 2bc \cos A.$$

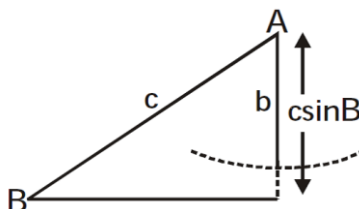
* If two sides b and c and an angle opposite the one of them (say B) are given then

$$\sin C = \frac{c}{b} \sin B, A = 180^\circ - (B + C) \text{ and } a = \frac{b \sin A}{\sin B} \text{ given the remaining elements.}$$

Case I :

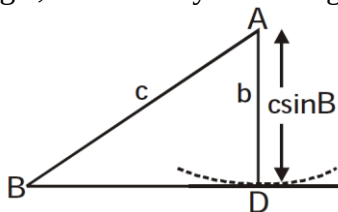
$$b < c \sin B.$$

We draw the side c and angle B . Now it is obvious from the figure that there is no triangle possible.



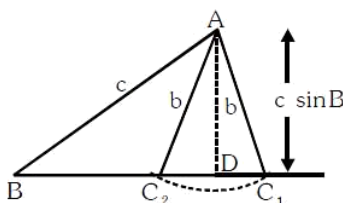
Case II :

$b = c \sin B$ and B is an acute angle, there is only one triangle possible. and it is right-angled at C .



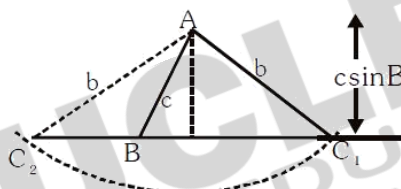
Case III :

$b > c \sin B$, $b < c$ and B is an acute angle, then there are two triangles possible for two values of angle C .



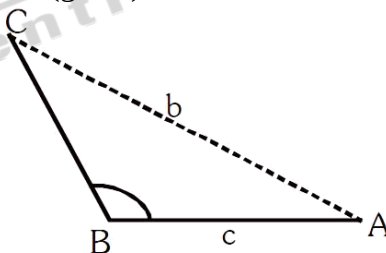
Case IV :

$b > c \sin B$, $c < b$ and B is an acute angle, then there is only one triangle.



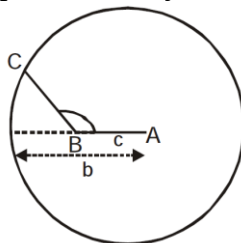
Case V :

$b > c \sin B$, $c > b$ and B is an obtuse angle. For any choice of point C , b will be greater than c which is a contradiction as $c > b$ (given). So there is no triangle possible.



Case VI :

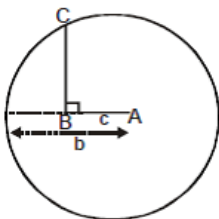
$b > c \sin B$, $c < b$ and B is an obtuse angle. We can see that the circle with A as centre and b as radius will cut the line only in one point. So only one triangle is possible.



Case VII :

$b > c$ and $B = 90^\circ$.

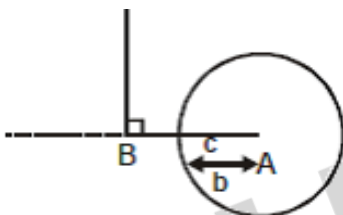
Again the circle with A as centre and b as radius will cut the line only in one point. So only one triangle is possible.



Case VIII :

$b \leq c$ and $B = 90^\circ$.

The circle with A as centre and b as radius will not cut the line in any point. So no triangle is possible.



This is, sometimes, called an ambiguous case.

Alternative Method :

By applying cosine rule, we have $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$$\Rightarrow a^2 - (2c \cos B)a + (c^2 - b^2) = 0$$

$$\Rightarrow a = c \cos B \pm \sqrt{(c \cos B)^2 - (c^2 - b^2)}$$

$$\Rightarrow a = c \cos B \pm \sqrt{b^2 - (c \sin B)^2}$$

This equation leads to following cases :

Case-I : If $b < c \sin B$, no such triangle is possible.

Case-II: Let $b = c \sin B$. There are further following case :

(a) B is an obtuse angle $\Rightarrow \cos B$ is negative. There exists no such triangle.

(b) B is an acute angle $\Rightarrow \cos B$ is positive. There exists only one such triangle.

Case-III: Let $b > c \sin B$. There are further following cases :

(a) B is an acute angle $\Rightarrow \cos B$ is positive. In this case triangle will exist if and only if $c \cos B > \sqrt{b^2 - (c \sin B)^2}$ or $c > b \Rightarrow$ Two such triangle is possible. If $c < b$, only one such triangle is possible.

(b) B is an obtuse angle $\Rightarrow \cos B$ is negative. In this case triangle will exist if and only if $\sqrt{b^2 - (c \sin B)^2} > |c \cos B| \Rightarrow b > c$. So in this case only one such triangle is possible. If $b < c$ there exists no such triangle.

This is called an ambiguous case.

* If one side a and angles B and C are given, then $A = 180^\circ - (B + C)$, and

$$b = \frac{a \sin B}{\sin A}, c = \frac{a \sin C}{\sin A}$$

* If the three angles A,B,C are given, we can only find the ratios of the sides a,b,c by using sine rule (since there are infinite similar triangles possible).

Illustration 15 : In the ambiguous case of the solution of triangles, prove that the circumcircles of the two triangles are of same size.

Solution : Let us say b,c and angle B are given in the ambiguous case. Both the triangles will have b and its opposite angle as B. so $\frac{b}{\sin B} = 2R$ will be given for both the triangles.

So their circumradii and therefore their sizes will be same.

Illustration 16 : If a,b and A are given in a triangle and c_1, c_2 are the possible values of the third side, prove that $c_1^2 + c_2^2 - 2c_1c_2 \cos 2A = 4a^2 \cos^2 A$.

Solution : $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

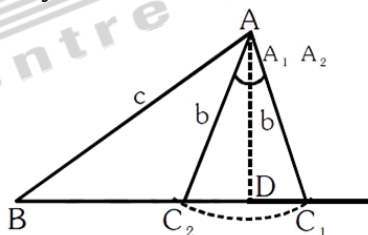
$$\Rightarrow c^2 - 2bc \cos A + b^2 - a^2 = 0.$$

$$c_1 + c_2 = 2b \cos A \text{ and } c_1 c_2 = b^2 - a^2.$$

$$\Rightarrow c_1^2 + c_2^2 - 2c_1c_2 \cos 2A = (c_1 + c_2)^2 - 2c_1c_2 (1 + \cos 2A) \\ = 4b^2 \cos^2 A - 2(b^2 - a^2)2 \cos^2 A = 4a^2 \cos^2 A.$$

Illustration 17 : If b, c, B are given and $b < c$, prove that $\cos \left(\frac{A_1 - A_2}{2} \right) = \frac{c \sin B}{b}$.

Solution : $\angle C_2AC_1$ is bisected by AD.



$$\Rightarrow \text{In } \triangle AC_2D, \cos \left(\frac{A_1 - A_2}{2} \right) = \frac{AD}{AC_2} = \frac{c \sin B}{b}$$

Hence proved.

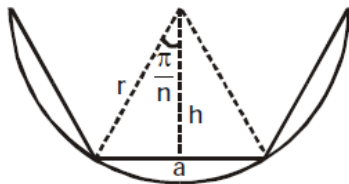
Do yourself - 9 :

(i) If b, c, B are given and $b < c$, prove that $\sin \left(\frac{A_1 - A_2}{2} \right) = \frac{a_1 - a_2}{2b}$

(ii) In a $\triangle ABC$, b, c, B ($c > b$) are given. If the third side has two values a_1 and a_2 such that $a_1 = 3a_2$, show that $\sin B = \sqrt{\frac{4b^2 - c^2}{3c^2}}$

14. REGULAR POLYGON :

A regular polygon has all its sides equal. It may be inscribed or circumscribed.

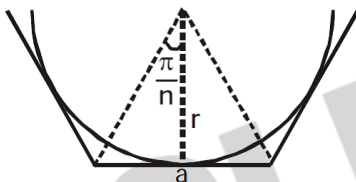


(a) Inscribed in circle of radius r :

(i) $a = 2h \tan \frac{\pi}{n} = 2r \sin \frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr \sin \frac{\pi}{n}$ and $A = \frac{1}{2} nr^2 \sin \frac{2\pi}{n}$

(b) Circumscribed about a circle of radius r :



(i) $a = 2r \tan \frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $P = 2nr \tan \frac{\pi}{n}$ and

$A = nr^2 \tan \frac{\pi}{n}$

Do yourself - 10 :

(i) If the perimeter of a circle and a regular polygon of n sides are equal, then prove that

$$\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$$

(ii) The ratio of the area of n -sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is 4 : 3. Find the value of n .

15. SOME NOTES :

(a) (i) If $a \cos B = b \cos A$, then the triangle is isosceles.

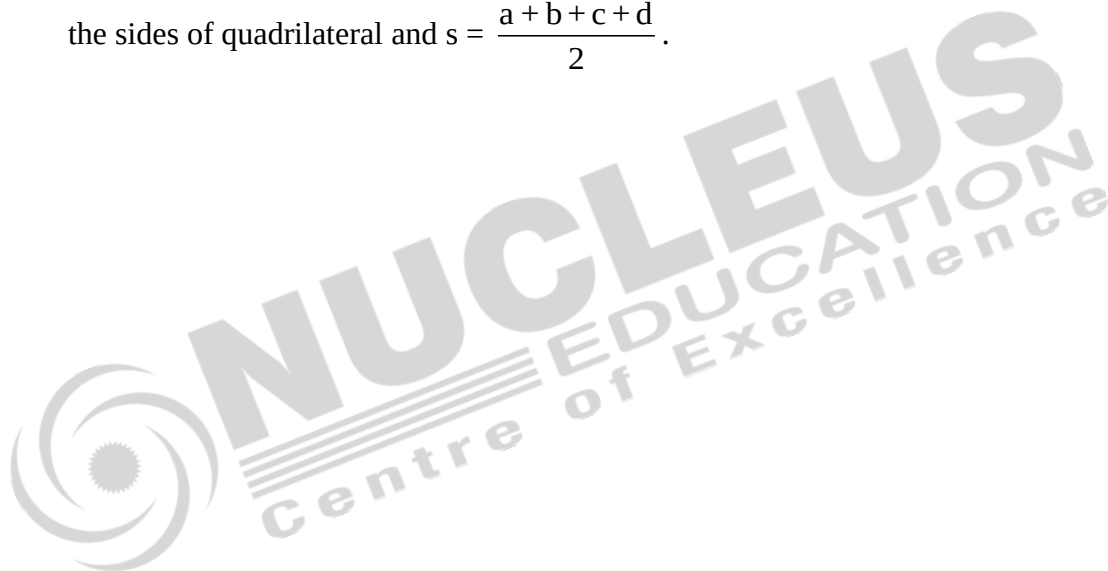
(ii) If $a \cos A = b \cos B$, then the triangle is isosceles or right angled.

(b) In right angle triangle

(i) $a^2 + b^2 + c^2 = 8R^2$ (ii) $\cos^2 A + \cos^2 B + \cos^2 C = 1$

(c) In equilateral triangle

- (i) $R = 2r$ (ii) $r_1 = r_2 = r_3 = \frac{3R}{2}$
- (iii) $r : R : r_1 = 1 : 2 : 3$ (iv) $\text{area} = \frac{\sqrt{3}a^2}{4}$ (v) $R = \frac{a}{\sqrt{3}}$
- (d) (i) The circumcentre lies (1) inside an acute angled triangle (2) outside an obtuse angled triangle & (3) mid point of the hypotenuse of right angled triangle.
 (ii) The orthocentre of right angled triangle is the vertex at the right angle.
 (iii) The orthocentre, centroid & circumcentre are collinear & centroid divides the line segment joining orthocentre & circumcentre internally in the ratio 2 : 1 except in case of equilateral triangle. In equilateral triangle, all these centres coincide
- (e) Area of a cyclic quadrilateral = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ where a, b, c, d are lengths of the sides of quadrilateral and $s = \frac{a+b+c+d}{2}$.


ANSWERS FOR DO YOURSELF

- | | | | | | | | |
|------------|------|-------------------|-------------------|---------------------------|---------------------------|-------------------|--------|
| 1 : | (i) | 90° | | | | | |
| 5 : | (i) | (a) $\frac{3}{5}$ | (b) $\frac{3}{4}$ | (c) $\frac{1}{\sqrt{10}}$ | (d) $\frac{3}{\sqrt{10}}$ | (e) $\frac{1}{3}$ | (f) 24 |
| 6 : | (i) | (a) 6 | (b) $\frac{5}{2}$ | (c) 1 | | | |
| 7 : | (i) | (a) 1 | (b) 3 | (c) $2\sqrt{3}$ | | | |
| 10: | (ii) | 6 | | | | | |

EXERCISE # 1 (A) ELEMENTARY

- Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$
- If K is a point on the side BC of an equilateral triangle ABC and if $\angle BAK = 15^\circ$, then the ratio of lengths $\frac{AK}{AB}$ is
 (A) $\frac{3\sqrt{2}(3+\sqrt{3})}{2}$ (B) $\frac{\sqrt{2}(3+\sqrt{3})}{2}$
 (C) $\frac{\sqrt{2}(3-\sqrt{3})}{2}$ (D) $\frac{3\sqrt{2}(3-\sqrt{3})}{2}$
- In a triangle ABC, $\angle A = 60^\circ$ and $b : c = (\sqrt{3} + 1) : 2$ then $(\angle B - \angle C)$ has the value equal to
 (A) 15° (B) 30° (C) 22.5° (D) 45°
- In an acute triangle ABC, $\angle ABC = 45^\circ$, $AB = 3$ and $AC = \sqrt{6}$. The angle $\angle BAC$, is
 (A) 60° (B) 65° (C) 75° (D) 15° or 75°
- In a triangle ABC, if $A = 30^\circ$ and $\frac{b}{c} = \frac{2+\sqrt{3}+\sqrt{2}-1}{2+\sqrt{3}-\sqrt{2}+1}$, then C, is equal to
 (A) 75° (B) 15° (C) $52\frac{1}{2}^\circ$ (D) $97\frac{1}{2}^\circ$
- In a triangle ABC, if $a = 6$, $b = 3$ and $\cos(A - B) = \frac{4}{5}$, the area of the triangle is
 (A) 8 (B) 9 (C) 12 (D) $\frac{15}{2}$
- In $\triangle ABC$, if $a = 2b$ and $A = 3B$, then the value of $\frac{c}{b}$ is equal to
 (A) 3 (B) $\sqrt{2}$ (C) 1 (D) $\sqrt{3}$
- If the sides of a triangle are $\sin \alpha$, $\cos \alpha$, $\sqrt{1 + \sin \alpha \cos \alpha}$, $0 < \alpha < \frac{\pi}{2}$, the largest angle is
 (A) 60° (B) 90° (C) 120° (D) 150°

9. If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of expression $E = \left(\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A \right)$, is
- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$
10. If in a triangle $\sin A : \sin C = \sin(A - B) : \sin(B - C)$ then $a^2 : b^2 : c^2$
- (A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these
11. In triangle ABC, if $\cot \frac{A}{2} = \frac{b+c}{a}$, then triangle ABC must be
- [Note: All symbols used have usual meaning in $\triangle ABC$.]
- (A) isosceles (B) equilateral (C) right angled (D) isosceles right angled
12. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If $a = 1$, $b = 3$ and $C = 60^\circ$, then $\sin^2 B$ is equal to
- (A) $\frac{27}{28}$ (B) $\frac{3}{28}$ (C) $\frac{81}{28}$ (D) $\frac{1}{3}$
13. The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. Then ratio of $A : B : C$ is
- (A) $3 : 5 : 2$ (B) $1 : \sqrt{3} : 2$ (C) $3 : 2 : 1$ (D) $1 : 2 : 3$
14. If the angles of a triangle are the ratio $4 : 1 : 1$, then the ratio of the longest side to the perimeter is
- (A) $\frac{2}{2+\sqrt{3}}$ (B) $\frac{2}{2-\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2+\sqrt{3}}$
15. In a triangle $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $a^2 : b^2 : c^2$ equals
- (A) $5 : 8 : 9$ (B) $5 : 8 : 12$ (C) $3 : 5 : 8$ (D) $5 : 8 : 10$
16. With 11, 13, $\sqrt{290 + 143\sqrt{3}}$ as sides
- (A) no triangle exists
- (B) triangle exists with an angle $\frac{2\pi}{3}$
- (C) triangle exists with an angle $\frac{3\pi}{4}$
- (D) triangle exists with an angle $\frac{5\pi}{6}$

17. In $\triangle ABC$ if $a = 8$, $b = 9$, $c = 10$, then the value of $\frac{\tan C}{\sin B}$ is
 (A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$
18. Given an acute triangle ABC such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and $AB = 50$. The area of the triangle ABC is equal to
 (A) 600 (B) 1200 (C) 1800 (D) 2400
19. In a triangle ABC , if a, b, c are in A.P. Then a possible value of B , is
 (A) 45° (B) 75° (C) 90° (D) 120°
20. If $\cos A + \cos B + 2 \cos C = 2$, then the sides of triangle ABC are in
 (A) A.P. (B) G.P. (C) H.P. (D) Not in A.P, G.P, H.P.
21. In a triangle ABC , if $\cos A \cos B + \sin A \sin B \sin C = 1$, then $a : b : c$ is equal to
 (A) $1:1:\sqrt{2}$ (B) $\sqrt{2}:\sqrt{3}:\sqrt{5}$ (C) $1:\sqrt{2}:1$ (D) $3:4:5$
22. In a triangle ABC , if the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum \left(\frac{\cos A}{a} \right) = \frac{p}{q}$, then find the least value of $(p + q)$.
23. In $\triangle ABC$, angle A is 120° , $BC + CA = 20$ and $AB + BC = 21$, then find the length of the side BC .
24. ABC is a triangle such that $\sin (2A + B) = \sin (C - A) = -\sin (B + 2C) = \frac{1}{2}$. If A, B, C are in A.P., find A, B, C .

EXERCISE # 1 (B)

- A triangle has vertices A, B and C, and the respective opposite sides have lengths a, b and c. This triangle is inscribed in a circle of radius R. If $b = c = 1$ and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then R equals
 (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2\sqrt{2}}$
- A circle is inscribed in a right triangle ABC, right angled at C. The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to
 (A) 91 (B) 96 (C) 100 (D) 104
- In a triangle ABC, if $a = 13$, $b = 14$ and $c = 15$, then angle A is equal to
 (All symbols used have their usual meaning in a triangle.)
 (A) $\sin^{-1} \frac{4}{5}$ (B) $\sin^{-1} \frac{3}{5}$ (C) $\sin^{-1} \frac{3}{4}$ (D) $\sin^{-1} \frac{2}{3}$
- In a triangle ABC, if $b = (\sqrt{3} - 1)a$ and $\angle C = 30^\circ$, then the value of $(A - B)$ is equal to
 (All symbols used have usual meaning in a triangle.)
 (A) 30° (B) 45° (C) 60° (D) 75°
- In triangle ABC, if $AC = 8$, $BC = 7$ and D lies between A and B such that $AD = 2$, $BD = 4$, then the length CD equals
 (A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$
- In a triangle ABC, if $\angle C = 105^\circ$, $\angle B = 45^\circ$ and length of side $AC = 2$ units, then the length of the side AB is equal to
 (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2} + 1$ (D) $\sqrt{3} + 1$
- In a triangle ABC, if $(a + b + c)(a + b - c)(b + c - a)(c + a - b) = \frac{8a^2b^2c^2}{a^2 + b^2 + c^2}$, then the triangle is
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) isosceles (B) right angled (C) equilateral (D) obtuse angled
- In triangle ABC, if $2b = a + c$ and $A - C = 90^\circ$, then $\sin B$ equals
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

9. In a triangle ABC, $a^3 + b^3 + c^3 = c^2(a + b + c)$
(All symbol used have usual meaning in a triangle.)
Statement-1 : The value of $\angle C = 60^\circ$.
Statement -2 : $\triangle ABC$ must be equilateral.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.
10. The sides of a triangle are three consecutive integers. The largest angle is twice the smallest one. The area of triangle is equal to
(A) $\frac{5}{4}\sqrt{7}$ (B) $\frac{15}{2}\sqrt{7}$ (C) $\frac{15}{4}\sqrt{7}$ (D) $5\sqrt{7}$
11. The sides a, b, c (taken in that order) of triangle ABC are in A.P.
If $\cos \alpha = \frac{a}{b+c}$, $\cos \beta = \frac{b}{c+a}$, $\cos \gamma = \frac{c}{a+b}$ then $\tan^2\left(\frac{\alpha}{2}\right) + \tan^2\left(\frac{\gamma}{2}\right)$ is equal to
[**Note**: All symbols used have usual meaning in triangle ABC.]
(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$
12. AD and BE are the medians of a triangle ABC. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$, $\angle ABE = \frac{\pi}{3}$, then area of triangle ABC equals
(A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{32}{9}\sqrt{3}$
13. In triangle ABC, if $\sin^3 A + \sin^3 B + \sin^3 C = 3\sin A \cdot \sin B \cdot \sin C$, then triangle is
(A) obtuse angled (B) right angled (C) obtuse right angled (D) equilateral
14. For right angled isosceles triangle, $\frac{r}{R} =$
[**Note**: All symbols used have usual meaning in triangle ABC.]
(A) $\tan \frac{\pi}{12}$ (B) $\cot \frac{\pi}{12}$ (C) $\tan \frac{\pi}{8}$ (D) $\cot \frac{\pi}{8}$
15. In triangle ABC, If $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$ then angle C is equal to
[**Note**: All symbols used have usual meaning in triangle ABC.]
(A) 30° (B) 45° (C) 60° (D) 90°
16. ABCD is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, then AB is equal to
[**JEE-MAIN 2013**]
(A) $\frac{(p^2 + q^2)\sin \theta}{p \cos \theta + q \sin \theta}$ (B) $\frac{p^2 + q^2 \cos \theta}{p \cos \theta + q \sin \theta}$ (C) $\frac{p^2 + q^2}{p^2 \cos \theta + q^2 \sin \theta}$ (D) $\frac{(p^2 + q^2)\sin \theta}{(p \cos \theta + q \sin \theta)^2}$

EXERCISE # 2

- Given a triangle ABC with sides $a = 7$, $b = 8$ and $c = 5$. If the value of the expression $(\sum \sin A)$
 $\left(\sum \cot \frac{A}{2}\right)$ can be expressed in the form $\frac{p}{q}$ where $p, q \in \mathbb{N}$ and $\frac{p}{q}$ is in its lowest form find the
value of $(p + q)$.
- If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the
squares of its sides then prove that the triangle is right angled.
- In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and
tangent to the other two sides. r_b and r_c are defined similarly. If r is the radius of the incircle of
triangle ABC then prove that, $\frac{2}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$.
- If the length of the perpendiculars from the vertices of a triangle A, B, C on the opposite sides
are p_1, p_2, p_3 then prove that $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$.
- With usual notations, prove that in a triangle ABC, $a \cot A + b \cot B + c \cot C = 2(R + r)$
- With usual notations, prove that in a triangle ABC, $Rr(\sin A + \sin B + \sin C) = \Delta$
- With usual notations, prove that in a triangle ABC, $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \frac{s^2}{\Delta}$
- With usual notations, prove that in a triangle ABC, $\cot A + \cot B + \cot C = \frac{a^2 + b^2 + c^2}{4\Delta}$
- If a, b, c are the sides of triangle ABC satisfying $\log\left(1 + \frac{c}{a}\right) + \log a - \log b = \log 2$.
Also $a(1 - x^2) + 2bx + c(1 + x^2) = 0$ has two equal roots. Find the value of $\sin A + \sin B + \sin C$.
- With usual notations, prove that in a triangle ABC, $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$
- With usual notations, prove that in a triangle ABC, $\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} = \frac{3}{r}$
- With usual notations, prove that in a triangle ABC, $\frac{abc}{s} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \Delta$
- With usual notations, prove that in a triangle ABC, $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$
- With usual notations, prove that in a triangle ABC, $2R \cos A = 2R + r - r_1$
- If $r_1 = r + r_2 + r_3$ then prove that the triangle is a right angled triangle.

EXERCISE # 3 (JM)

1. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is : **[JEE (Main) - 2019]**
 (1) 5 : 6 : 7 (2) 3 : 4 : 5 (3) 4 : 5 : 6 (4) 5 : 9 : 13
2. The angles A, B and C of a triangle ABC are in A.P. and $a : b = 1 : \sqrt{3}$. If $c = 4$ cm, then the area (in sq. cm) of this triangle is : **[JEE (Main) - 2019]**
 (1) $4\sqrt{3}$ (2) $\frac{2}{\sqrt{3}}$ (3) $2\sqrt{3}$ (4) $\frac{4}{\sqrt{3}}$
3. If 5, $5r$, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to: **[JEE (Main) - 2019]**
 (1) $\frac{7}{4}$ (2) $\frac{5}{4}$ (3) $\frac{3}{2}$ (4) $\frac{3}{4}$
4. With the usual notation in $\triangle ABC$, if $\angle A + \angle B = 120^\circ$, $a = \sqrt{3} + 1$ and $b = \sqrt{3} - 1$, then the ratio $\angle A : \angle B$, is : **[JEE (Main) - 2019]**
 (1) 5 : 3 (2) 9 : 7 (3) 3 : 1 (4) 7 : 1
5. Given $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ for a $\triangle ABC$ with usual notation. If $\frac{\cos A}{\alpha} = \frac{\cos B}{\beta} = \frac{\cos C}{\gamma}$, then the ordered triad (α, β, γ) has a value: **[JEE (Main) - 2019]**
 (1) (3, 4, 5) (2) (5, 12, 13) (3) (7, 19, 25) (4) (19, 7, 25)

EXERCISE # 4 (JA)

- The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. The ratio $A : B : C$ is
[JEE 2004 (Screening)]
(A) $3 : 5 : 2$ (B) $1 : \sqrt{3} : 2$ (C) $3 : 2 : 1$ (D) $1 : 2 : 3$
- (a) In $\triangle ABC$, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is
[JEE 2005 (Screening)]
(A) $(b - c) \sin\left(\frac{B - C}{2}\right) = a \cos\left(\frac{A}{2}\right)$
(B) $(b - c) \cos\left(\frac{A}{2}\right) = a \sin\left(\frac{B - C}{2}\right)$
(C) $(b + c) \sin\left(\frac{B + C}{2}\right) = a \cos\left(\frac{A}{2}\right)$
(D) $(b - c) \cos\left(\frac{A}{2}\right) = 2a \sin\left(\frac{B + C}{2}\right)$
(b) Circles with radii 3, 4 and 5 touch each other externally if P is the point of intersection of tangents to these circles at their points of contact. Find the distance of P from the points of contact.
[JEE 2005 (Mains), 2]
- (a) Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is
[JEE 2006, 3]
(A) $7 + 12\sqrt{3}$ (B) $12 - 7\sqrt{3}$ (C) $12 + 7\sqrt{3}$ (D) 4π
(b) Internal bisector of $\angle A$ of a triangle ABC meets side BC at D. A line drawn through D perpendicular to AD intersects the side AC at E and the side AB at F. If a, b, c represent sides of $\triangle ABC$ then
(A) AE is HM of b and c (B) $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$
(C) $EF = \frac{4bc}{b+c} \sin \frac{A}{2}$ (D) the triangle AEF is isosceles [JEE 2006, 5]
- Let ABC and ABC' be two non-congruent triangles with sides $AB = 4$, $AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is
[JEE 2009, 5]
- (a) If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the length of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$, is -
(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

- (b) Consider a triangle ABC and let a, b and c denote the length of the sides opposite to vertices A, B and C respectively. Suppose $a = 6$, $b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to
- (c) Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is/are [JEE 2010, 3+3+3]
- (A) $-(2 + \sqrt{3})$ (B) $1 + \sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $4\sqrt{3}$
6. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals [JEE 2012, 3M, -1M]
- (A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$
7. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are) [JEE(Advanced) 2013, 3, (-1)]
- (A) 16 (B) 18 (C) 24 (D) 22
8. In a triangle the sum of two sides is x and the product of the same two sides is y . If $x^2 - c^2 = y$, where c is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is - [JEE(Advanced)-2014, 3(-1)]
- (A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$
9. In a triangle XYZ, let x, y, z be the lengths of sides opposite to the angles X, Y, Z respectively and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then [JEE (Advanced) 2016]
- (A) Area of the triangle XYZ is $6\sqrt{6}$
- (B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$
- (C) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$
- (D) $\sin^2 \left(\frac{X+Y}{2} \right) = \frac{3}{5}$

10. In a triangle PQR let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively, Then, which of the following statement(s) is are TRUE ?
[JEE (Advanced) 2018, 4/3/2/1(-2)]
- (A) $\angle QPR = 45^\circ$
- (B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120^\circ$
- (C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$
- (D) The area of the circumcircle of the triangle PQR is 100π .
11. In a non-right angled triangle ΔPQR , let p, q, r denote the lengths of the sides opposite to the angles at P, Q, R respectively. The median from R meets the side PQ at S, then perpendicular from P meets the side QR and E, and RS and PE intersect at O. If $p = \sqrt{3}$, $q = 1$ and the radius of the circumcircle of the ΔPQR equals 1, then which of the following options is/are correct ?
[JEE Advanced - 2019]
- (A) Length of OE = $\frac{1}{6}$
- (B) Length of RS = $\frac{\sqrt{7}}{2}$
- (C) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}}{2}(2 - \sqrt{3})$
- (D) Area of $\Delta SOE = \frac{\sqrt{3}}{12}$

EXERCISE # 5

- In a triangle ABC, if $\angle A = 30^\circ$, $b = 10$ and $a = x$, then the values of x for which there are 2 possible triangles is given by (All symbols used have usual meaning in a triangle.)
(A) $5 < x < 10$ (B) $x < \frac{5}{2}$ (C) $\frac{5}{3} < x < 10$ (D) $\frac{5}{2} < x < 10$
- Let ABC be a right triangle with length of side $AB = 3$ and hypotenuse $AC = 5$. If D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is equal to
(A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$
- If the lengths of the medians of a triangle ABC are 6, 8, 10 then area of triangle is equal to
(A) 24 (B) 32 (C) 16 (D) 36
- In triangle ABC, If $s = 3 + \sqrt{3} + \sqrt{2}$, $3B - C = 30^\circ$, $A + 2B = 120^\circ$, then the length of longest side of triangle is
[Note: All symbols used have usual meaning in triangle ABC.]
(A) 2 (B) $2\sqrt{2}$ (C) $2(\sqrt{3} + 1)$ (D) $\sqrt{3} - 1$
- In $\triangle ABC$, if a, b, c (taken in that order) are in A.P. then $\cot \frac{A}{2} \cot \frac{C}{2} =$
[Note: All symbols used have usual meaning in triangle ABC.]
(A) 1 (B) 2 (C) 3 (D) 4
- In triangle ABC, if $\Delta = a^2 - (b - c)^2$, then $\tan A =$
[Note: All symbols used have usual meaning in triangle ABC.]
(A) $\frac{15}{16}$ (B) $\frac{1}{2}$ (C) $\frac{8}{17}$ (D) $\frac{8}{15}$

Multiple Correct Answer Type :

- In a triangle ABC, let $2a^2 + 4b^2 + c^2 = 2a(2b + c)$, then which of the following holds good?
[Note: All symbols used have usual meaning in a triangle.]
(A) $\cos B = \frac{-7}{8}$ (B) $\sin(A - C) = 0$
(C) $\frac{r}{r_1} = \frac{1}{5}$ (D) $\sin A : \sin B : \sin C = 1 : 2 : 1$

8. In a triangle ABC, if $a = 4$, $b = 8$, $\angle C = 60^\circ$, then which of the following relations is (are) correct?
[**Note:** All symbols used have usual meaning in triangle ABC.]
- (A) The area of triangle ABC is $8\sqrt{3}$
(B) The value of $\sum \sin^2 A = 2$
(C) Inradius of triangle ABC is $\frac{2\sqrt{3}}{3+\sqrt{3}}$
(D) The length of internal angle bisector of angle C is $\frac{4}{\sqrt{3}}$
9. In which of the following situations, it is possible to have a triangle ABC?
(All symbols used have usual meaning in a triangle.)
- (A) $(a + c - b)(a - c + b) = 4bc$ (B) $b^2 \sin 2C + c^2 \sin 2B = ab$
(C) $a = 3$, $b = 5$, $c = 7$ and $C = \frac{2\pi}{3}$ (D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$
10. In a triangle ABC, which of the following quantities denote the area of the triangle?
- (A) $\frac{a^2 - b^2}{2} \left(\frac{\sin A \sin B}{\sin(A-B)} \right)$ (B) $\frac{r_1 r_2 r_3}{\sqrt{\sum r_1 r_2}}$
(C) $\frac{a^2 + b^2 + c^2}{\cot A + \cot B + \cot C}$ (D) $r^2 \cot \frac{A}{2} \cdot \cot \frac{B}{2} \cot \frac{C}{2}$
11. In $\triangle ABC$, angle A, B and C are in the ratio $1 : 2 : 3$, then which of the following is (are) correct?
(All symbol used have usual meaning in a triangle.)
- (A) Circumradius of $\triangle ABC = c$ (B) $a : b : c = 1 : \sqrt{3} : 2$
(C) Perimeter of $\triangle ABC = 3 + \sqrt{3}$ (D) Area of $\triangle ABC = \frac{\sqrt{3}}{8} c^2$
12. Let one angle of a triangle be 60° , the area of triangle is $10\sqrt{3}$ and perimeter is 20 cm. If $a > b > c$ where a, b and c denote lengths of sides opposite to vertices A, B and C respectively, then which of the following is (are) correct?
- (A) Inradius of triangle is $\sqrt{3}$ (B) Length of longest side of triangle is 7
(C) Circumradius of triangle is $\frac{7}{\sqrt{3}}$ (D) Radius of largest escribed circle is $\frac{1}{12}$

13. In triangle ABC, let $b = 10$, $c = 10\sqrt{2}$ and $R = 5\sqrt{2}$ then which of the following statement(s) is(are) correct?
[**Note:** All symbols used have usual meaning in triangle ABC.]
(A) Area of triangle ABC is 50.
(B) Distance between orthocentre and circumcentre is $5\sqrt{2}$
(C) Sum of circumradius and inradius of triangle ABC is equal to 10
(D) Length of internal angle bisector of $\angle ACB$ of triangle ABC is $\frac{5}{2\sqrt{2}}$
14. In a triangle ABC, let $BC = 1$, $AC = 2$ and measure of angle C is 30° . Which of the following statement(s) is (are) correct?
(A) $2 \sin A = \sin B$
(B) Length of side AB equals $5 - 2\sqrt{3}$
(C) Measure of angle A is less than 30°
(D) Circumradius of triangle ABC is equal to length of side AB
15. Let P be an interior point of $\triangle ABC$.
Match the correct entries for the ratios of the Area of $\triangle PBC$: Area of $\triangle PCA$: Area of $\triangle PAB$ depending on the position of the point P w.r.t. $\triangle ABC$.
- | Column-I | Column-II |
|-----------------------------|-----------------------------------|
| (A) If P is centroid (G) | (P) $\tan A : \tan B : \tan C$ |
| (B) If P is incentre (I) | (Q) $\sin 2A : \sin 2B : \sin 2C$ |
| (C) If P is orthocentre (H) | (R) $\sin A : \sin B : \sin C$ |
| (D) If P is circumcentre | (S) $1 : 1 : 1$ |
| | (T) $\cos A : \cos B : \cos C$ |

EXERCISE # 6

- With usual notation, if in a ΔABC , $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$; then prove that,

$$\frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}.$$
- Given a triangle ABC with $AB = 2$ and $AC = 1$. Internal bisector of $\angle BAC$ intersects BC at D . If $AD = BD$ and Δ is the area of triangle ABC, then find the value of $12\Delta^2$.
- For any triangle ABC, if $B = 3C$, show that $\cos C = \sqrt{\frac{b+c}{4c}}$ and $\sin \frac{A}{2} = \frac{b-c}{2c}$.
- In a triangle ABC if $a^2 + b^2 = 101c^2$ then find the value of $\frac{\cot C}{\cot A + \cot B}$.
- The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is 60° . If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.
- If in a ΔABC , $a = 6$, $b = 3$ and $\cos(A-B) = 4/5$ then find its area.
- In a ΔABC , (i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$
 (iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$, prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).
- Two sides of a triangle are of lengths $\sqrt{6}$ and 4 and the angle opposite to smaller side is 30° . How many such triangles are possible? Find the length of their third side and area.
- The triangle ABC (with side lengths a, b, c as usual) satisfies $\log a^2 = \log b^2 + \log c^2 - \log (2bc \cos A)$. What can you say about this triangle?
- The sides of a triangle are consecutive integers $n, n+1$ and $n+2$ and the largest angle is twice the smallest angle. Find n .

ANSWER KEY
EXERCISE # 1 (A) ELEMENTARY

- | | | | | | | |
|--------|--------|------------------------------------|-------|-------|-------|-------|
| 1. C | 2. C | 3. B | 4. C | 5. C | 6. B | 7. D |
| 8. C | 9. D | 10. A | 11. C | 12. A | 13. D | 14. D |
| 15. A | 16. D | 17. A | 18. B | 19. A | 20. A | 21. A |
| 22. 25 | 23. 13 | 24. $45^\circ, 60^\circ, 75^\circ$ | | | | |

EXERCISE # 1 (B)

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. A | 3. A | 4. C | 5. C | 6. D | 7. B |
| 8. C | 9. C | 10. C | 11. D | 12. D | 13. D | 14. C |
| 15. C | 16. A | | | | | |

EXERCISE # 2

- | | |
|--------|-------------------|
| 1. 107 | 9. $\frac{12}{5}$ |
|--------|-------------------|

EXERCISE # 3 (JM)

- | | | | | |
|------|------|------|------|------|
| 1. 3 | 2. 3 | 3. 1 | 4. 4 | 5. 3 |
|------|------|------|------|------|

EXERCISE # 4 (JA)

- | | | | | | |
|----------|----------|----------------|-----------|-------------|------|
| 1. D | 2.(a) B; | (b) $\sqrt{5}$ | 3. (a) C, | (b) A,B,C,D | 4. 4 |
| 5.(a) D, | (b) 3, | (c) B | 6. C | 7. B,D | 8. B |
| 9. ACD | 10. BCD | 11. ABC | | | |

EXERCISE # 5

- | | | | | | | |
|--------------------------------|--------|-----------|---------|---------|-----------|-----------|
| 1. A | 2. B | 3. B | 4. C | 5. C | 6. D | 7. B,C |
| 8. A,B | 9. B,C | 10. A,B,D | 11. B,D | 12. A,C | 13. A,B,C | 14. A,C,D |
| 15. (A) S; (B) R; (C) P; (D) Q | | | | | | |

EXERCISE # 6

- | | | | |
|---|-------|------------------|---------------|
| 2. 9 | 4. 50 | 5. 3 cms & 2 cms | 6. 9 sq. unit |
| 8. Two triangle $(2\sqrt{3}-\sqrt{2})$, $(2\sqrt{3}+\sqrt{2})$, $(2\sqrt{3}-\sqrt{2})$, and $(2\sqrt{3}+\sqrt{2})$ sq. units | | | |
| 9. triangle is isosceles | 10. 4 | | |

Result JEE Main

Result Highlights 2019



100 Percentile
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2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



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Shreya Pathak



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Abhay Goyal



Piyush Tibarwal



Abhinav Kumar



Tushar Gautam



Saurav Sahoo



Pratyush

Mentored by Team



Kushal Babel
2014



Gourika Bang
2015



Amit Malav
2013



Suraj Sanjay Jog
2012



Chitraang Murdia
2014



Pulkit Tandon
2012



Kapil Shobhnani
2016



Utkarsh Sinha
2015



S John S
2015